Richardson's Extrapolation

Richardson's Extrapolation

- Richardson's Extrapolation improves accuracy of numerical approximations.
  - Uses two step sizes (h and h/2).
  - Eliminates leading error term.

Error

- If approximation d(h) has error of form:
  - $E(h) = c \cdot h^p$
  - Then $D = d(h) + c \cdot h^p$

Richardson's Formula

- When h2 = h/2:

$$D \approx \frac{2^p\, d(h/2) - d(h)}{2^p - 1}$$

Example: Derivative of e^x at x = 0

- Using forward difference with h = 0.1:
  - d(h) = 1.05170
  - d(h/2) = 1.02542
  - Richardson: $D \approx 2 \cdot d(h/2) - d(h) = 0.99914$
  - True value: p=1

Richardson Extrapolation

- Enhances accuracy significantly.
- Cancels leading error term.
- Useful for derivatives, integrals, ODEs.

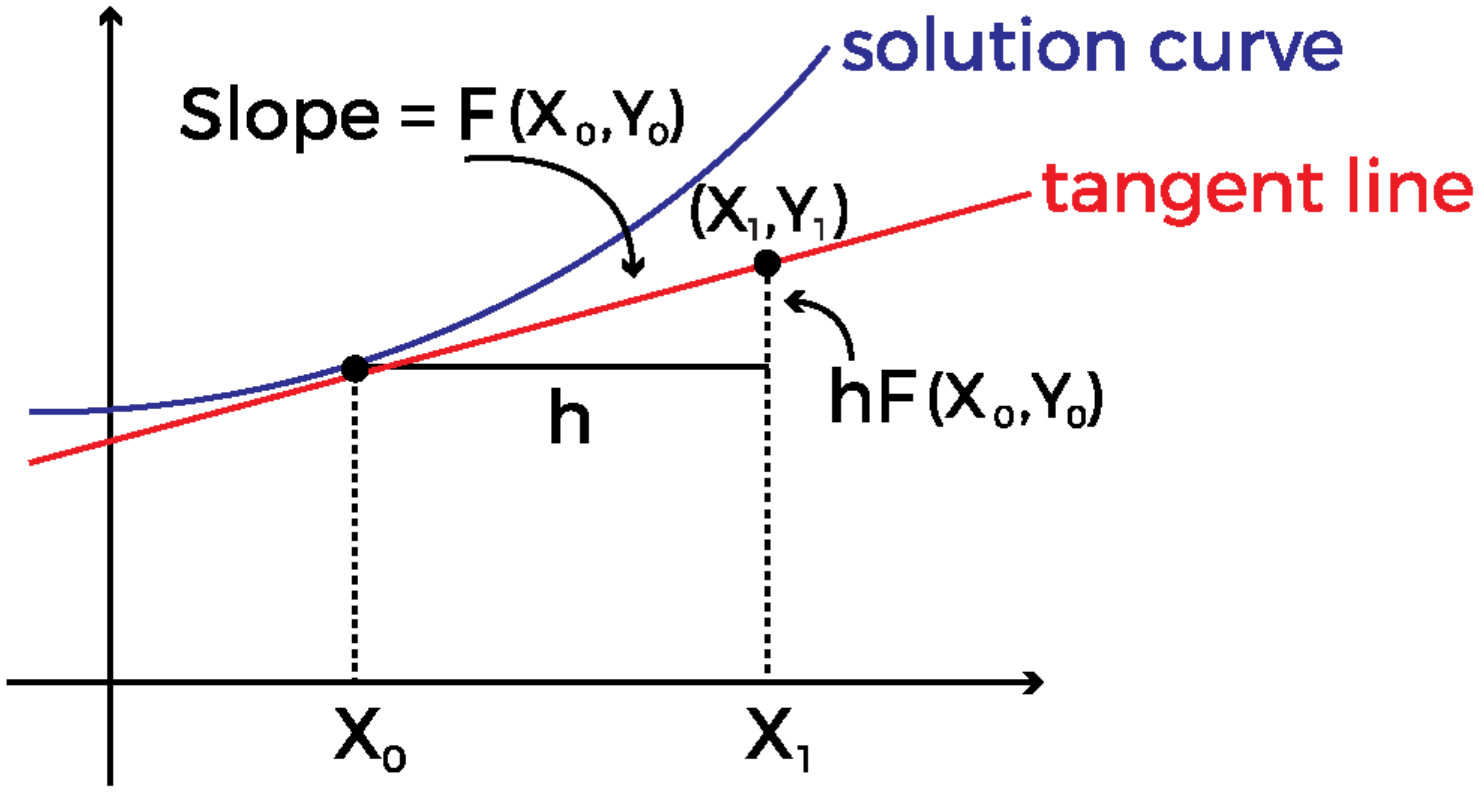
Euler Method

Euler Method

Given $f'(t) = \frac{dy}{dt} = f(t, y)$

Euler Method uses the slope at a point to estimate the next value

$$y_{n+1} = y_n + h\, f(t_n, y_n)$$


solution curve
Slope = F(X0,Y0)
tangent line
(X1,Y1)
hF(X0,Y0)
h
X0
X1

Euler Method

The differential equation

$$\frac{dy}{dt} = f(t, y)$$

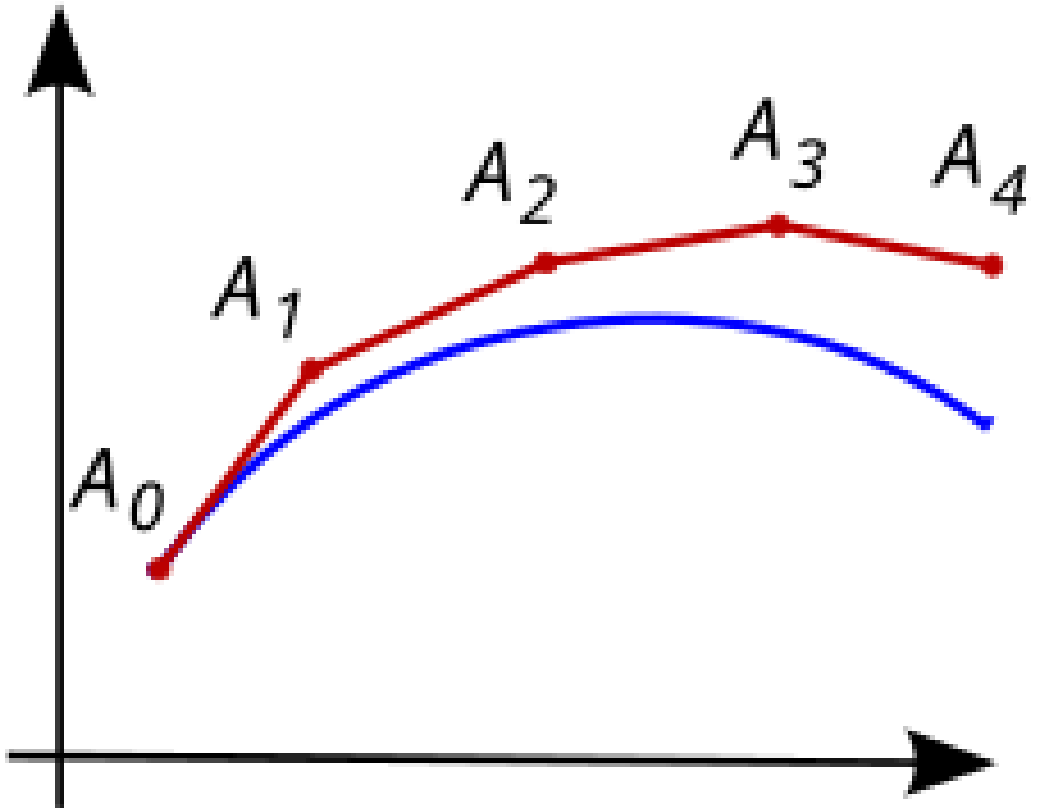
tells us the **slope** of the solution at any point.

If we know the slope at (t_n, y_n), then for a small step h:

$$y(t_{n+1}) \approx y(t_n) + \text{slope} \times \text{step}.$$

Euler's formula:

$$\boxed{y_{n+1} = y_n + h\, f(t_n, y_n)}$$

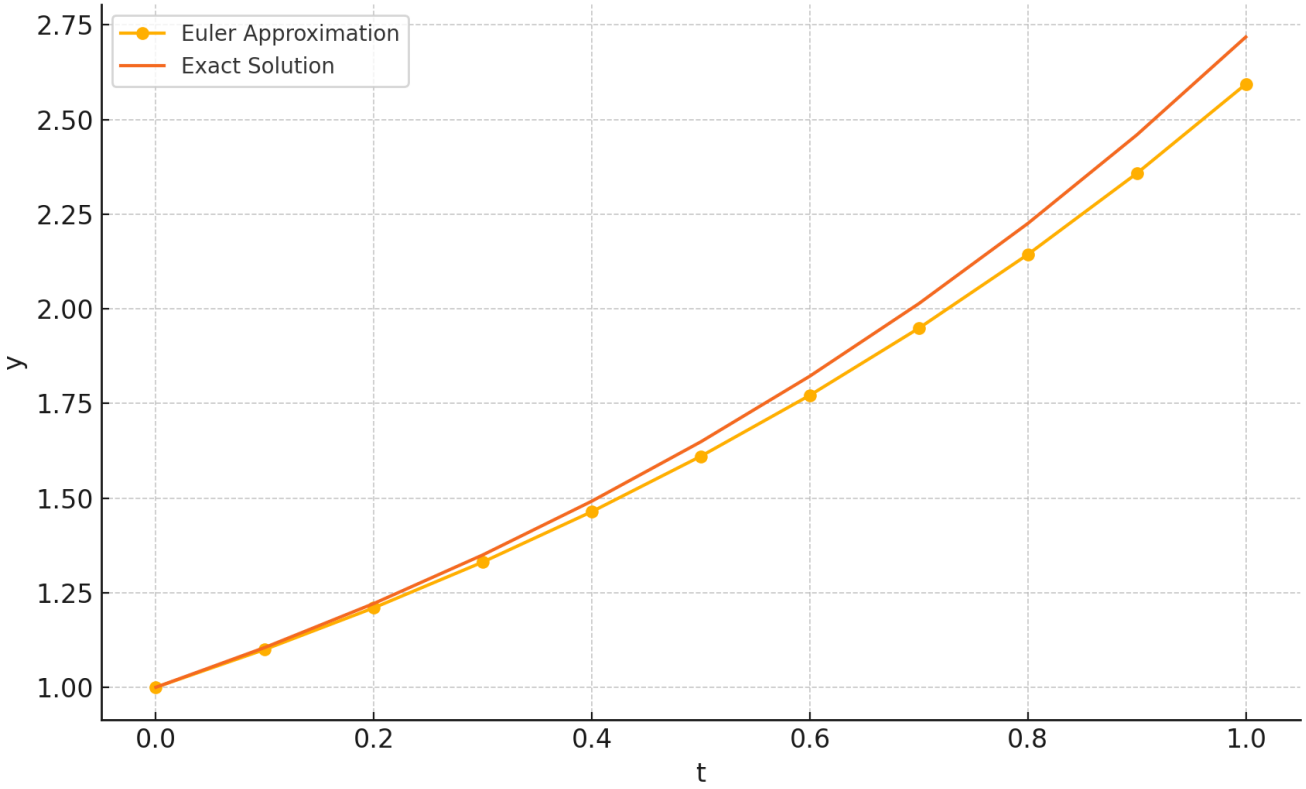

A0
A1
A2
A3
A4

Example

$$y(t) = e^t$$

Using step size $h = 0.1$, find $y(0.1)$

Approximation VS Exact



Error Analysis In Euler's Method

Actual solution of the differential equation is

$$y = \frac{x^2}{2} + 1$$

n	x_n	y_n (actual)	y_n (approx)	% error
0	0	1	1	0
1	0.1	1.005	1	0.497512
2	0.2	1.02	1.01	0.980392
3	0.3	1.045	1.03	1.435407
4	0.4	1.08	1.06	1.851852

Euler's Method

Euler's Method uses the slope *only at the beginning* of each interval:

$$y_{n+1}^{(Euler)} = y_n + hf(t_n, y_n).$$

This can be inaccurate because the slope often changes between t_n and t_{n+1}.

Improved Euler's Method

Given (t_n, y_n):

Step 1: Predictor (Euler step)

Compute a temporary value:

$$y_n^* = y_n + h\, f(t_n, y_n).$$

Step 2: Corrector

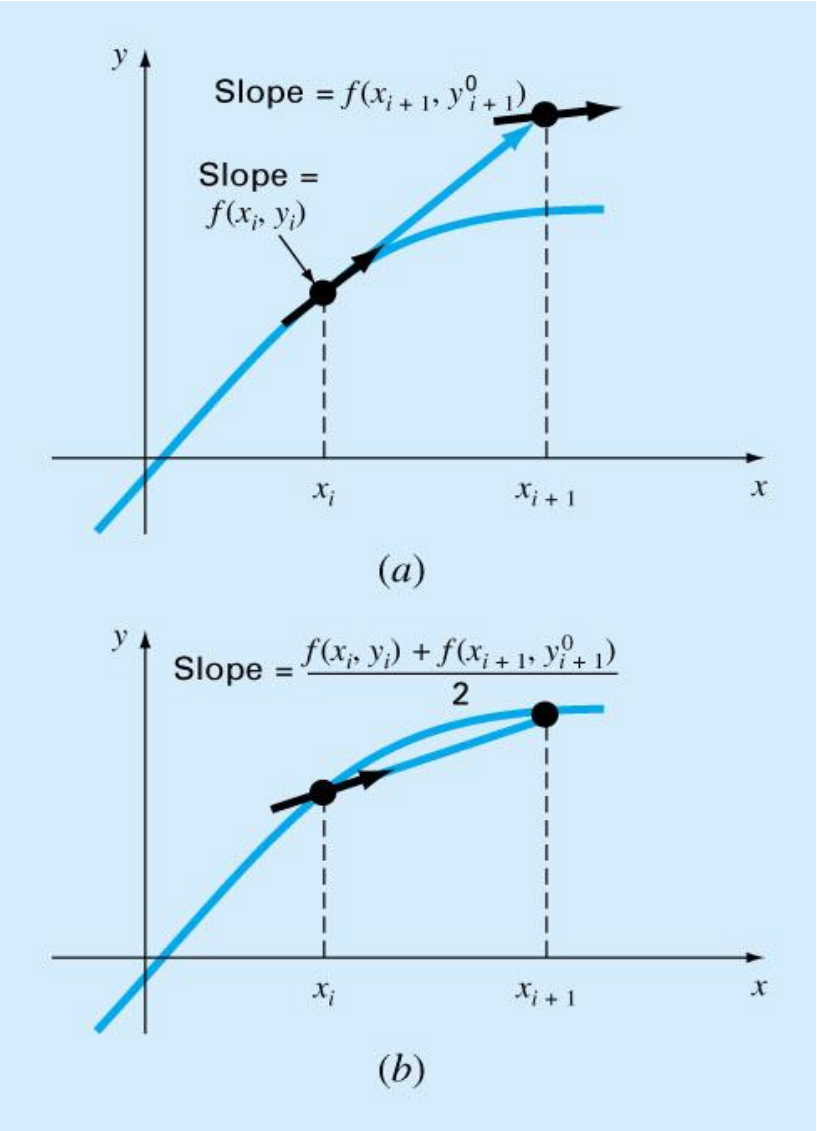
Evaluate slope at the predicted point:

$$f^* = f(t_n + h, y_n^*).$$

Step 3: Take the average slope

$$y_{n+1} = y_n + \frac{h}{2}\left[f(t_n, y_n) + f^*\right]$$

Improved Euler's Method



Improved Euler's Method

$$k_1 = f(t_n, y_n),$$
$$k_2 = f(t_n + h,\ y_n + hk_1),$$
$$y_{n+1} = y_n + \frac{h}{2}(k_1 + k_2).$$

Second Order form of RK Method

Runge-Kutta Method

Fourth Order Runge-Kutta

Compute slope at four places within each step

$$k_1 = f(t_j, y_j)$$
$$k_2 = f(t_j + \frac{h}{2}, y_j + \frac{h}{2}k_1)$$
$$k_3 = f(t_j + \frac{h}{2}, y_j + \frac{h}{2}k_2)$$
$$k_4 = f(t_j + h, y_j + hk_3)$$

Use weighted average of slopes to obtain y_{j+1}

$$y_{j+1} = y_j + h\left(\frac{k_1}{6} + \frac{k_2}{3} + \frac{k_3}{3} + \frac{k_4}{6}\right)$$

FIGURE 25.15
Graphical depiction of the slope estimates comprising the fourth-order RK method.

Fourth Order Runge-Kutta

Compute slope at four places within each step

$$k_1 = f(t_j, y_j)$$

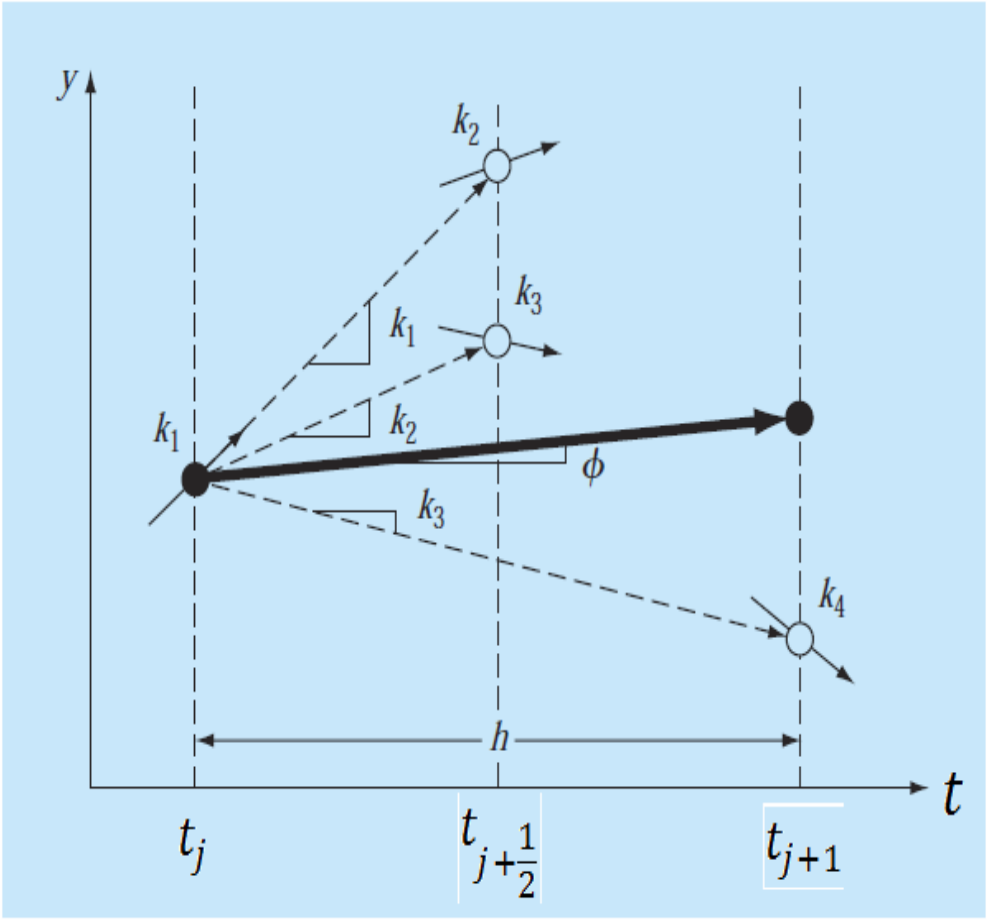
$$k_2 = f\left(t_j + \frac{h}{2}, y_j + \frac{h}{2}k_1\right)$$

$$k_3 = f\left(t_j + \frac{h}{2}, y_j + \frac{h}{2}k_2\right)$$

$$k_4 = f(t_j + h, y_j + hk_3)$$

Use weighted average of slopes to obtain y_{j+1}

$$y_{j+1} = y_j + h\left(\frac{k_1}{6} + \frac{k_2}{3} + \frac{k_3}{3} + \frac{k_4}{6}\right)$$



3rd Order Runge-Kutta

Know as RK3

$$k_1 = f(x_i, y_i)$$

$$k_2 = f\left(x_i + \frac{h}{2},\ y_i + \frac{1}{2}k_1 h\right)$$

$$k_3 = f(x_i + h,\ y_i - k_1 h + 2k_2 h)$$

$$y_{i+1} = y_i + \frac{h}{6}\left(k_1 + 4k_2 + k_3\right)$$

Local error is $O(h^4)$ and Global error is $O(h^3)$

4th Order Runge-Kutta

$$k_1 = f(x_i, y_i)$$

$$k_2 = f(x_i + \frac{h}{2}, \quad y_i + \frac{1}{2}k_1 h)$$

$$k_3 = f(x_i + \frac{h}{2}, \quad y_{\mathrm{i}} + \frac{1}{2}k_2 h)$$

$$k_4 = f(x_i + h, \quad y_{\mathrm{i}} + k_3 h)$$

$$y_{i+1} = y_i + \frac{h}{6}\left(k_1 + 2k_2 + 2k_3 + k_4\right)$$

$$\text{Local error is } O(h^5) \text{ and global error is } O(h^4)$$

Higher-Order Runge-Kutta

$$k_1 = f(x_i, y_i)$$

$$k_2 = f(x_i + \frac{1}{4}h,\ y_i + \frac{1}{4}k_1 h)$$

$$k_3 = f(x_i + \frac{1}{4}h,\ y_i + \frac{1}{8}k_1 h + \frac{1}{8}k_2 h)$$

$$k_4 = f(x_i + \frac{1}{2}h,\ y_i - \frac{1}{2}k_2 h + k_3 h)$$

$$k_5 = f(x_i + \frac{3}{4}h,\ y_i + \frac{3}{16}k_1 h + \frac{9}{16}k_4 h)$$

$$k_6 = f(x_i + h,\ y_i - \frac{3}{7}k_1 h + \frac{2}{7}k_2 h + \frac{12}{7}k_3 h - \frac{12}{7}k_4 h + \frac{8}{7}k_5 h)$$

$$y_{i+1} = y_i + \frac{h}{90}\left(7k_1 + 32k_3 + 12k_4 + 32k_5 + 7k_6\right)$$

Example

4th-Order Runge-Kutta Method

$$\frac{dy}{dx} = 1 + y + x^2$$

$$y(0) = 0.5$$

$$h = 0.2$$

$$Use\ RK4\ to\ compute\ \ y(0.2)\ and\ \ y(0.4)$$

Summary

- Runge Kutta methods generate an accurate solution without the need to calculate high order derivatives.
- Second order RK have local truncation error of order $O(h^3)$ and global truncation error of order $O(h^2)$.
- Higher order RK have better local and global truncation errors.
- N function evaluations are needed in the Nth order RK method.